MILITARY SPECIFICATION

PIN, BONE, BAYONET POINT, KIRSCHNER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bone pins, with bayonet points, of the Kirschner design, for applying traction in the extension of fractures.

1.2 Classification. Bone pins shall be of the following sizes and styles as specified (see 6.2):

Size	Length (inches)
1	9
2	12

Style	Diameter (inch)
1	0.035
2	0.045
3	0.062

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications
PPP-B-566	Boxes, Folding Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.

FSC 6515

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Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151 Metals; Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

2055 PIN, BONE, BAYONET POINT, KIRSCHNER

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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ASTM F86-68 Recommended Practice for Surface Requirements of Metallic Surgical Implants.
ASTM F55-66 Stainless Steel Bars and Wire for Surgical Implants.
ASTM E45-63 Determining the Inclusion Content of Steel.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENT

3.1 Material. The bone pins shall be fabricated from cold drawn, spring temper wire, conforming to the composition specified in ASTM F55-66 concerning stainless steel material used for surgical implants, except that the carbon content shall be 0.03 percent maximum. Tensile strength of wire shall be 220,000 pounds per square inch when tested as specified in 4.4.2. Wire shall not crack, tear, or break when tested as specified in 4.4.4.

3.1.1 Inclusion content of material. The nonmetallic inclusion content of the raw material (steel billet) shall be determined by the JERNKONTORET (JK) chart method in accordance with ASTM E45-63, using method "A or D." The numbers listed in Table I shall represent the maximum allowable inclusion ratings.

Table I. Maximum Allowable Inclusion Ratings In Accordance with ASTM E45

	Type A	Type B	Type C	Type D
Thin Series	2	1-1/2	1-1/2	2
Thick Series	1-1/2	1	1	1

3.2 Hardness. After proper stress relieving, Rockwell hardness of the cutting portions of the bone pins shall be C40 minimum when tested as specified in 4.4.3.

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3.3 Finish. The completed bone pin shall have a bright, smooth, mirror finish, properly electropolished. The shaped ends of the bone drill shall have a mirror or satin finish as described and specified in GG-I-526. The finish shall be free from all surface imperfections. Definitions of acceptable surface characteristics shall be as described in ASTM F86.

3.3.1 Passivation. The bone pins shall be properly passivated after completion of all final finishing operations and shall be accomplished by the methods described in ASTM F86.

3.4 Style, design and dimensions. Shall be a one-piece bone pin with bayonet point, with or without triangular shank end. Shall be of the Kirschner design, suitable for implanting into the body for applying traction in the extension of fractured bone segments. The bone pins shall be suitable for use with FSN 6515-310-7800 BONE SURGERY SET, Extension and Traction, Pin Type, Steinmann and FSN 6515-312-3500 DRILL, HAND BONE, Smedberg. The bone pins shall be in accordance with DPSC Drawing 2055, for appropriate size, stock number, and item identification.

3.5 Resistance to boil test, copper sulfate test and saline test. Bone pins shall not show evidence of corrosion when subjected to the boil test as specified in 4.4.1.1; the copper sulfate test as specified in 4.4.1.2, and the saline test as specified in 4.4.1.3.

3.6 Workmanship. Workmanship shall be first class throughout. Bone pins shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality supplied by the manufacturer of the cold drawn, spring temper austenitic steel wire may be furnished in lieu of actual performance of such testing by the contractor, provided lot identify has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II.

Table II. Sampling Criteria for Examination

Type Defect	Inspection Level *	AQL(percent defective)
Major	S-4	1.0
Minor	S-2	2.5

* If acceptance number if not exceeded, lot shall be accepted. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III.

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Table III. Sampling Criteria for Tests

Component	Characteristic	Req.	Test Procedure	Unit of Product	Insp. level	AQL
Bone Pin	Tensile Strength	3.1	4.4.2	1-pin	S-2	1.5
	Bend Test	3.1	4.4.4	1-pin	S-2	1.5
	Inclusion Content	3.1.1	ASTM E45	1-pin*	S-1	1.0
	Hardness	3.2	4.4.3	1-pin	S-2	1.5
	Dimensions	3.4	- - -	1-pin	S-2	1.5
	Boil Test	3.5	4.4.1.1	1-pin	S-4	0.65
	Copper Sulfate Test	3.5	4.4.1.2	1-pin	S-4	0.65
	Saline Test	3.5	4.4.1.3	1-pin	S-4	0.65

* The unit lot of steel, for which the inclusion content is to be determined, shall not be larger than one heat.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in construction of the bone pins shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bone pins shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Inspection, however, shall not be restricted to defects listed in Table IV. Any bone pins in the sample containing one or more defects shall be rejected and if the number of defects in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Bone pins not of finish specified.
102	Bone pin does not have sharp bayonet point.
103	Bone pins not free from cracks, pits, scale, slivers and tool marks.
Minor	
201	Bayonet point of bone pins not protected in unit package.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Bone pins shall be subjected to the boil test as described and specified in Federal Specification GG-I-526.

4.4.1.2 Copper Sulfate test. Bone pins shall be subjected to the copper sulfate test as described and specified in Federal Specification GG-I-526.

4.4.1.3 Saline test. Bone pins shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Bone pins shall then be completely immersed into Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 ± 1 degrees Centigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Tensile strength test. Tensile strength of the bone pins shall be conducted in accordance with Test Method 211.1 of Federal Test Method Standard No. 151.

4.4.3 Hardness test. Rockwell hardness of the bone pins shall be determined in accordance with method 243 of Federal Test Method Standard No. 151.

4.4.4 Bend test. The bone pins shall be bent 180 degrees over a 1/8 inch diameter mandrel and shall be noted for deformation, tearing, cracking or breakage.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (12 pins) as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-8.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4. Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3,

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style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bone pins are intended for implanting into the body for applying traction in the extension of fractured bone segments.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Size and style required (see 1.2).

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers items appearing in the Federal Supply Catalog as indicated below. It does not include all styles and sizes as indicated by the title nor all types commercially available.

Federal Stock Number	Item Identification
6515-310-9280	PIN, BONE, BAYONET POINT, KIRSCHNER, 9 by 0.035 inch, 12s
6515-310-9300	PIN, BONE, BAYONET POINT, KIRSCHNER, 9 by 0.045 inch, 12s
6515-310-9320	PIN, BONE, BAYONET POINT, KIRSCHNER, 9 by 0.062 inch, 12s
6515-310-9340	PIN, BONE, BAYONET POINT, KIRSCHNER, 12 by 0.035 inch, 12s
6515-310-9360	PIN, BONE, BAYONET POINT, KIRSCHNER, 12 by 0.045 inch, 12s
6515-310-9380	PIN, BONE, BAYONET POINT, KIRSCHNER, 12 by 0.062 inch, 12s

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6.4 Supersession data This specification includes requirements of Purchase Description No. 8, dated 4 March 1969.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM
Project No. 6515-1459

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-36634 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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